

Abdelhamid F. Farhat

Cairo, Egypt

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EDUCATION

Arab Academy for Science, Technology & Maritime Transport (AASTMT)

Aswan, Egypt

B.S. in Computer Science

Feb 2023 - June 2026

- **GPA:** 3.70/4.00
- **Related Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Organization & Programming, Operating Systems, Software Engineering, Introduction to Artificial Intelligence, Advanced Artificial Intelligence, Deep Learning, Machine Learning, Statistics & Applications, Web Development, Problem Solving.

PUBLICATION

- Farhat, A., et al. "SAOE-YOLO11m: An Efficient Small Aerial Objects Expert Detection Model." IEEE ICEENG IICC 2026.

RESEARCH EXPERIENCE

Computer Vision Research

Cairo, Egypt

Co-author, IEEE ICEENG IICC 2026

Sep 2025 – May 2026

- Co-authored "SAOE-YOLO11m: An Efficient Small Aerial Objects Expert Detection Model," accepted and presented at the IEEE ICEENG IICC 2026 conference, focused on improving small aerial object detection using an optimized YOLOv11m architecture.
- Executed the full experimental pipeline including dataset preparation, model training, evaluation against baseline YOLO variants, and ablation studies to optimize detection accuracy on low-resolution aerial imagery.

PROFESSIONAL EXPERIENCE

Freelance Web Developer

Remote | [Live](#)

Freelance Developer

Jan 2026 – Present

- Delivered a full-stack literary portfolio and content management system for author Hossam Abdelmalek using React, Vite, and Supabase, enabling centralized management of poems, books, articles, and audio recitations through a custom CMS.
- Developed interactive user-focused features including downloadable social-media poem cards using the HTML5 Canvas API, a custom audio player, and Progressive Web App (PWA) support to improve accessibility and user engagement.
- Designed and implemented scalable full-stack software systems with emphasis on performance, usability, and maintainability, including UI/UX design, database architecture, deployment, optimization, and production monitoring on Vercel.

PROJECTS

ASRDS Ground Control Station

Aswan, Egypt | [Source Code](#)

Graduation Project — Team Member

Sep 2025 – May 2026

- Developed a JavaFX-based Ground Control Station (GCS) for the Autonomous Search and Rescue Drone System (ASRDS), enabling real-time communication with autonomous drones through serial connectivity, live telemetry streaming, and mission-area transmission for search-and-rescue operations.
- Engineered an interactive monitoring and control interface featuring real-time GPS tracking, flight-path visualization, battery and altitude monitoring, mission polygon configuration, and bidirectional serial communication using jSerialComm, Gluon Maps, and custom telemetry protocols.
- Implemented both software simulation and real-world drone deployment workflows, including built-in drone simulation, Python-based serial simulators, virtual COM communication testing, and successful integration with a physical autonomous drone platform for live field operation and mission execution.

Dabbarha

Aswan, Egypt | [live](#)

Solo Developer

Feb 2026 – Present

- Developed a cross-platform personal finance platform for MENA users consisting of a responsive web application and a React Native mobile app, using Next.js, React Native, Supabase, Clerk Authentication, and Recharts to deliver real-time financial dashboards and spending analytics.

- Engineered gamification and AI-powered financial features including achievement systems, smart notifications, automated expense categorization, and personalized financial insights using Google Generative AI to improve user engagement and financial awareness.
- Implemented production-grade infrastructure including transactional email workflows with Resend, security hardening with Arcjet, responsive data visualizations, and scalable backend integrations to ensure reliability, accessibility, and seamless synchronization across web and mobile platforms.

FarhaLink

Team Member

Aswan, Egypt | [Live](#)

Apr 2026 – May 2026

- Built a production-grade SaaS platform enabling users to create customizable digital wedding invitations with RSVP management, QR-based guest validation, and integrated gift catalogs using Next.js, React, TypeScript, and Supabase.
- Engineered scalable performance features including Incremental Static Regeneration (ISR), optimized caching strategies, and a custom client-side QR scanner to support high-traffic event access and real-time RSVP verification.
- Designed a fully RTL Arabic-first user experience tailored for the MENA market, integrating WhatsApp sharing, Google Maps location parsing, and calendar event generation to improve accessibility and user engagement.

GetBox

Solo Developer

Aswan, Egypt | [Source Code](#)

Jan 2026 – Apr 2026

- Built an open-source media downloading platform using Next.js and Vercel Edge Functions, supporting video, audio, and image extraction from YouTube, Instagram, TikTok, X/Twitter, Reddit, SoundCloud, and other platforms without third-party paid APIs.
- Engineered a globally optimized edge-first architecture with static generation, immutable asset caching, lightweight dependencies, and sequential fallback extractors to deliver fast response times and reliable media resolution at scale.
- Developed a mobile-first Progressive Web App (PWA) featuring installability, keyboard shortcuts, URL history, SEO optimization, security hardening, and direct media link detection to improve usability and cross-platform accessibility.

CERTIFICATIONS & TRAINING

IEEE Student Branch – Aswan University

Aswan, Egypt

1st Place — Build with AI Competition

Apr 2026

- Secured 1st place in an AI-focused hackathon organized by the IEEE Student Branch by developing and presenting an AI-powered application from concept to working demo within the competition timeline.

Mahara-Tech / ITI

Remote | [Preview](#)

React JS Certificate

Aug 2025

- Completed React JS training covering component-based architecture, state management, and modern frontend development practices.

Sprints × Microsoft Summer Camp

Remote | [Preview](#)

AI & Machine Learning Track

Jul 2025

- Completed a 40-hour training program covering AI and machine learning fundamentals, workflows, and practical model development concepts.

Sprints × Microsoft Summer Camp

Remote | [Preview](#)

Web Development Track

Jun 2025

- Completed a 40-hour intensive training program focused on modern web development fundamentals and frontend engineering practices.

Information Technology Institute (ITI)

Remote | [Preview](#)

Full Stack Web Development using PHP

Sep 2024

- Completed a 120-hour full-stack web development training program covering PHP, MySQL, Laravel, and modern web technologies.

SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, PHP, SQL, C, C++

Frontend & Mobile: React.js, React Native, Next.js, HTML/CSS, Vite

Backend & APIs: Node.js, Express.js

Databases & Backend Services: Supabase, Firebase

AI/ML: Machine Learning, Deep Learning, Google Generative AI

Desktop & Systems: JavaFX, Serial Communication

Tools & DevOps: Git, Vercel, Arcjet, Clerk Authentication, Resend